

Let f be a C^1 -function of the real axis in any case

$$f(x) = \frac{1}{x^2} \exp(-x^2) \quad (1)$$

Give the Laplace transform $\mathcal{L}f$ of f as a function of s and show that $\mathcal{L}f$ is integrable.

$$\mathcal{L}f(s) = \frac{1}{2} \frac{1}{s^2} \exp(-s^2) \quad (2)$$

where

$$\mathcal{L}f(s) = \int_0^\infty f(x) e^{-sx} dx = \int_0^\infty \frac{1}{x^2} \exp(-x^2) e^{-sx} dx$$

The integral in (2) is convergent as

$$\frac{1}{x^2} \exp(-x^2) \leq \frac{1}{x^2} \quad (3)$$

Since $\frac{1}{x^2}$ is not integrable on $(0, \infty)$, we need to use the fact that $\exp(-x^2)$ decays faster than $\frac{1}{x^2}$ as $x \rightarrow \infty$.

By the dominated convergence theorem, we can interchange the order of integration in (2) and get

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Theorem 1. Let f be a function defined on $[a, b]$ and let F be an antiderivative of f . Then $F(b) - F(a) = \int_a^b f(x) dx$.

Proof. The function F is continuous on $[a, b]$ and differentiable on (a, b) . By the Mean Value Theorem, there exists a point c in (a, b) such that $F(b) - F(a) = (b - a)F'(c)$. Since $F'(c) = f(c)$, we have $F(b) - F(a) = (b - a)f(c)$.

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